

NEXASAT DIGITAL WALLET

Customer Lifetime Value (LTV) Prediction

Executive Summary Report

Prepared for Executive Leadership
Data Science & Analytics
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1. Executive Summary

NexaSat Digital Wallet undertook a structured data science initiative to predict Customer Lifetime Value (LTV) across its 7,000-customer base, to identify which customers are most valuable and what drives that value. Eight modelling approaches were built and rigorously tested, culminating in a tuned CatBoost model selected for production use.

- The final model explains approximately 74% of the variation in customer LTV on data it had never seen before and generalises well (small gap between training and test performance).
- Two behavioural metrics shows how much a customer spends per active day, and their average transaction value are overwhelmingly the strongest predictors of LTV, together accounting for most of the model's predictive power.
- Demographic and firmographic factors (ie. location, income level, preferred payment method, app usage frequency) show only marginal (2 to 4%) differences in average LTV between groups, these factors carry comparatively little weight in determining customer value.
- Simpler, interpretable models (Decision Tree) performed nearly as well as more complex ones; added complexity (Random Forest, XGBoost, Neural Network) did not translate into materially better, more reliable predictions, and in several cases overfit the training data.
- Recommendation: LTV growth and retention strategy should prioritise deepening customer engagement (transaction frequency and basket size) over demographic-based segmentation.

2. Business Objective

Understanding which customers will generate the most long-term value allows NexaSat to prioritise retention spend, personalise engagement, and forecast revenue more accurately. This project set out to build a reliable, production-ready model that predicts individual customer LTV from behavioural, engagement and demographic data, and to surface the key drivers of that value for strategic decision-making.

3. Data & Analytical Approach

The analysis was built on a clean, high-quality dataset of 7,000 customers with no missing values. Redundant, highly correlated engineered features (e.g. spend-per-transaction, which duplicated average transaction value) were identified and removed to keep the model stable and avoid double-counting the same underlying signal. Remaining features were checked for multicollinearity (Variance Inflation Factor analysis) and confirmed low risk. All numerical features were standardised prior to modelling.

Customers analysed	7,000
Predictor variables used	15 (after removing redundant and correlated features)
Missing values	0 (fully resolved prior to modelling)
Train / test split	5,600 training and 1,400 independent test (80/20)
Validation method	5-fold cross-validation throughout
Average customer LTV	511,920 (median 387,818 distribution is right skewed)

Table 1: Dataset and validation summary

4. Model Development & Selection

A staged approach was used: simple, interpretable baseline models were built first, followed by more advanced machine-learning and deep-learning approaches, all evaluated consistently using an independent 20% hold-out test set and 5-fold cross-validation. This ensures reported performance reflects how each model would behave on genuinely new customers, not just the data it was trained on.

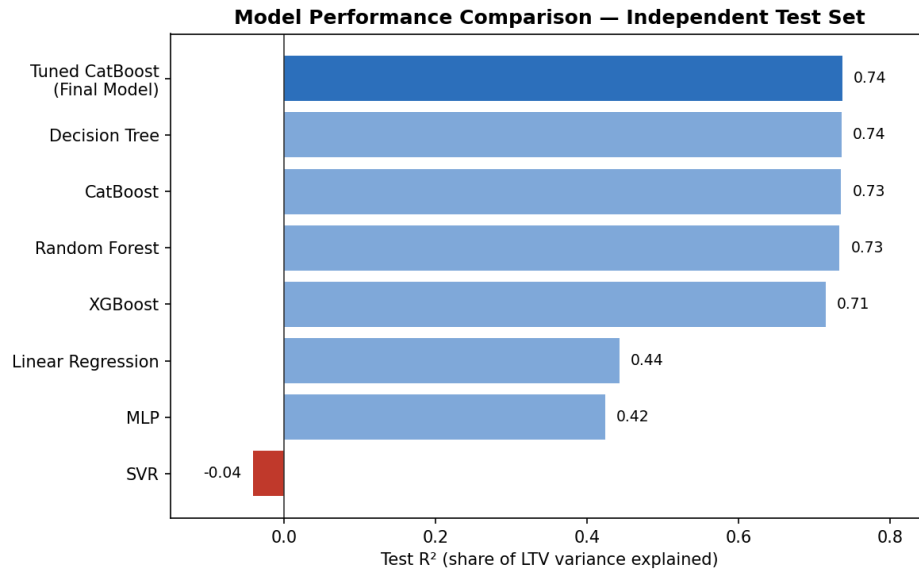


Figure 1: Test-set R^2 across all models evaluated. Higher is better; models overfitting to training data (Random Forest, XGBoost) or under-fitting (SVR, Neural Network) were not selected.

Model	Stage	Test R^2	Test RMSE	Test MAE
Linear Regression	Baseline	0.44	329,543	243,560
Decision Tree	Baseline	0.74	226,900	151,245
Random Forest	Advanced	0.73	228,100	152,170
XGBoost	Advanced	0.71	235,673	155,766
CatBoost (default)	Advanced	0.73	227,257	154,427
Neural Network (MLP)	Advanced	0.42	335,000	250,653
Support Vector Regression	Advanced	-0.04	450,229	340,844
Tuned CatBoost (FINAL)	Selected	0.74	226,407	152,767

Table 2: Full model comparison on the independent test set

The Decision Tree and CatBoost family of models clearly outperformed linear regression, support vector regression, and the neural network (MLP), confirming that LTV is driven by non-linear relationships and interactions the simpler and deep-learning approaches did not capture as effectively here. Random Forest and XGBoost achieved strong training performance but showed a large gap versus test performance (22 percentage points), indicating overfitting. CatBoost, after hyperparameter tuning (RandomizedSearchCV, 5-fold cross-validation, 250 model fits), delivered the best balance of accuracy and generalisation and was selected as the final production model.

5. Final Model Performance

The selected Tuned CatBoost model was evaluated on 1,400 customers it had never seen during training or tuning:

Metric	Value	Plain-English Meaning
Test R ²	0.737	Explains 73.7% of customer-to-customer LTV variation
Test MAE	152,767	Typical prediction is off by 152.8K on average
Test RMSE	226,407	Larger errors are penalised more; confirms no extreme miscalibration
Test MAPE	55.8%	Best used for ranking/tiering customers, not exact value forecasts
Train–Test R ² Gap	0.028	Small gap confirms the model generalises well to new customers

Table 3: Final model performance, in business terms

In practical terms, the model is reliable for ranking and segmenting customers by expected value (e.g. identifying the top 10 to 20% highest-LTV customers for premium retention treatment), but individual predicted LTV values should be treated as directional rather than exact, given the 56% average percentage error.

6. What Drives Customer Lifetime Value

Feature importance was assessed using SHAP (Shapley Additive Explanations), the industry-standard method for explaining individual predictions from complex models. The results are striking: two engagement metrics dominate all other factors combined.

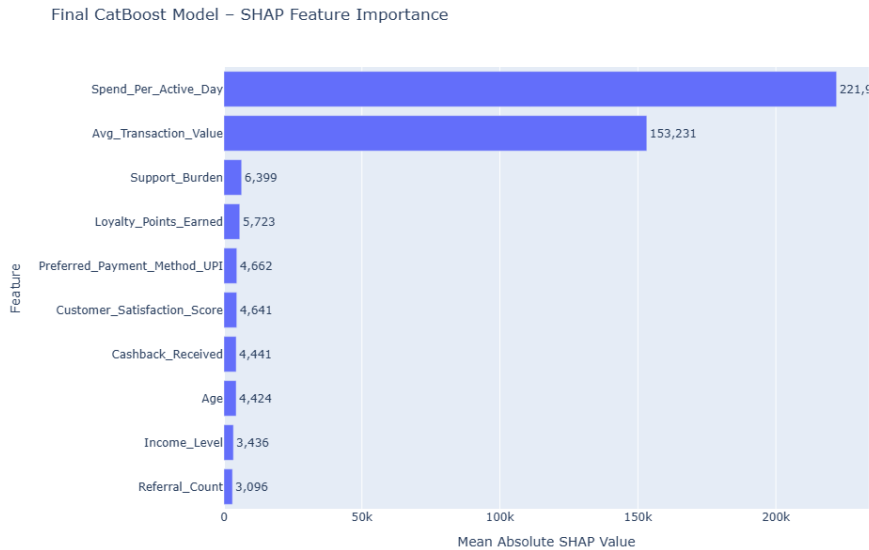


Figure 2: SHAP feature importance for the final CatBoost model (higher = greater influence on predicted LTV)

- Spend per active day is, by a wide margin, the single biggest driver of predicted LTV is roughly 30 times more influential than most other features.
- Average transaction value is the second-strongest driver, reinforcing that transaction depth (not just frequency) matters.

- All remaining factors support tickets raised, loyalty points, payment method, satisfaction score, cashback, age, income level, referrals each contribute only a small, roughly comparable amount individually.

This pattern indicates LTV is primarily an outcome of how actively and deeply a customer transacts, rather than who they are demographically.

7. Segment-Level Insights

Average LTV was compared across the main customer segments captured in the dataset. Differences between groups are consistently small, typically within 2 to 4% of one another reinforcing the SHAP finding that demographic and channel segments are not meaningful differentiators of customer value on their own.

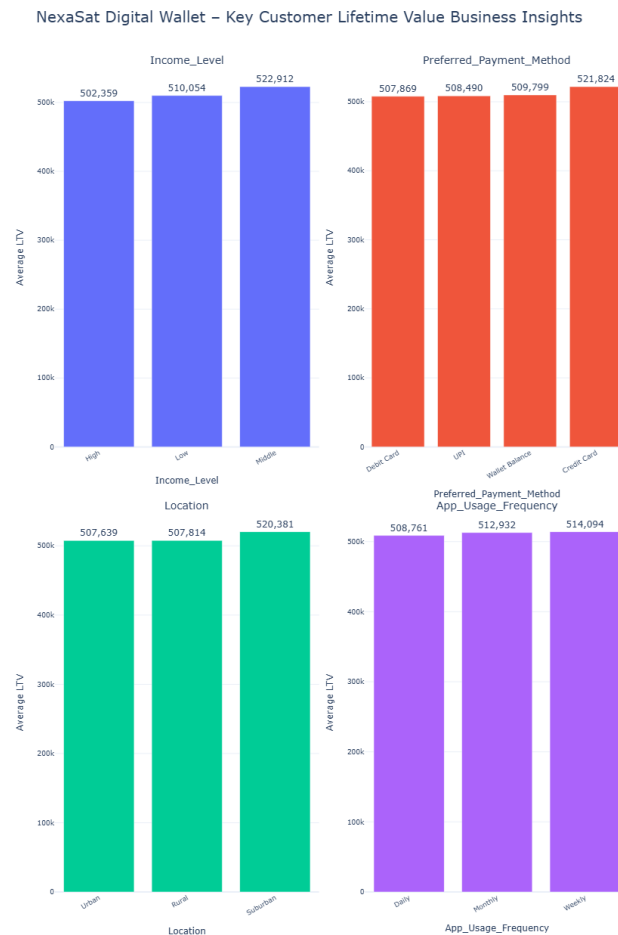


Figure 3: Average LTV by income level, preferred payment method, location, and app usage frequency

- Income level: Middle-income customers show marginally higher average LTV (522.9K) than High (502.4K) or Low (510.1K) and a difference too small to justify income-based LTV targeting alone.
- Location: Suburban customers show slightly higher average LTV (520.4K) than Urban (507.6K) or Rural (507.8K).
- Payment method and app usage frequency: differences of under 3% across all categories with essentially no practical effect on LTV.

8. Business Recommendations

- Shift retention and growth investment toward engagement, not demographics. Since spend-per-active-day and average transaction value dominate LTV, prioritise initiatives that increase how often and how much customers transact (e.g. habit-forming nudges, bundled offers, cashback timed to lapsing activity) over demographic-based campaign targeting.
- Build an early-warning signal on declining engagement. Because spend-per-active-day is the top predictor, a meaningful drop in this metric is likely the earliest sign of falling future LTV, it recommended to flag it for proactive retention outreach before churn risk becomes visible in other metrics.
- Use the model to rank and tier customers, not to forecast exact revenue. Deploy the Tuned CatBoost model to score all customers and segment them into value tiers (e.g. top 10%, next 20%, etc.) for differentiated treatment, this is where the model adds the most reliable business value.
- Reduce reliance on income and location-based personalisation specifically for LTV strategy. These factors showed negligible differentiation in this analysis; segmentation budgets are likely better spent on behavioural triggers.
- Establish a periodic retraining cadence. Customer behaviour evolves; refresh the model on a regular cycle (e.g. quarterly) using the same validated pipeline to keep predictions current.

9. Limitations & Next Steps

- Prediction precision: an average percentage error of 56% means the model is well-suited to relative ranking and tiering but should not be relied upon for precise individual revenue forecasts.
- Unexplained variation: roughly 26% of LTV variation remains unexplained by the current feature set; incorporating additional data such as acquisition channel, tenure, product mix, or macroeconomic indicators could improve accuracy further.
- Skewed target variable: LTV is highly right-skewed, with a small number of very high-value customers. Consider a dedicated high-value customer model or outlier-aware treatment in a future iteration.
- Point-in-time snapshot: the model reflects a single snapshot of customer behaviour; ongoing monitoring and retraining will be required to maintain accuracy as the business and customer base evolve.

Overall, this project delivers a validated, production-ready LTV model and a clear, actionable finding for the business: customer value is earned through engagement, not defined by demographics.